

Pilot 01 Report: Controlled Pilot Stress Maps

Validated Baseline American CN/PSOR Solver

Codex-assisted pilot diagnostics report

June 17, 2026

Abstract

This report documents the first controlled pilot stress-map experiment after the solver validation ladder and downstream design gate. The pilot uses the baseline American Crank–Nicolson / projected SOR solver only. It produces small diagnostic tables and figures for an American put family and a dividend-paying American call family. The outputs are pilot diagnostics, not production risk surfaces, training labels, broad parameter sweeps, or neural surrogate inputs.

Contents

1 Purpose of the Pilot	3
2 Link to Prior Gates	3
3 Solver and Grid Choice	3
4 Pilot Case Definitions	3
5 Output Definitions	4
6 Diagnostic Requirements	4
7 Tables and Interpretation	4
8 Figures and Interpretation	5
9 Acceptance Criteria Results	7
10 Limitations	8
11 What This Pilot Supports	8
12 What This Pilot Does Not Support	8
13 Recommended Next Step	8

14 Reproducibility Record

9

1 Purpose of the Pilot

Ticket 12 concluded `PASS_SOLVER_VALIDATION_WITH_LIMITATIONS`. The downstream design gate then approved a small, human-reviewed pilot before any large-scale stress maps or data generation. This Pilot 01 report records that first narrow experiment.

The purpose is practical: verify that the validated classical solver can produce interpretable stress-map style figures together with the metadata needed to read them safely. The pilot keeps the scope deliberately small. It checks option values, payoffs, continuation premium, exercise/continuation indicators, approximate boundary curves, PSOR convergence, LCP diagnostics, and masked Greek diagnostics.

2 Link to Prior Gates

The pilot follows the validation path documented in the Ticket 12 synthesis report and the downstream design gate. The theory basis remains the classical Black–Scholes PDE and American finite-difference/LCP setting (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995). The Rannacher comparison from Ticket 10A remains a comparison tool only (Rannacher, 1984); it is not used as the pilot default.

Gate decision	Pilot implementation rule
Use validated baseline solver	All runs use <code>american_crank_nicolson_psor_price</code> .
Keep Rannacher optional	No Pilot 01 run uses the Rannacher solver.
Use normalized strike	Every run uses $K = 1$.
Use practical pilot grid	Main runs use $S_{\max} = 4$, $M = N = 120$.
Include higher-grid check	Base put and base dividend call also run at $M = N = 180$.
Keep outputs diagnostic	Manifest status is <code>pilot_diagnostic_only</code> .

3 Solver and Grid Choice

The default solver is the Ticket 04 baseline American CN/PSOR solver. The pilot uses

$$S_{\max} = 4K, \quad M = N = 120$$

for the main cases and $M = N = 180$ for the two higher-grid checks. The main interpretation region is $S/K \in [0.4, 1.8]$. The solver still computes on the full domain because the boundary conditions and diagnostics depend on the full grid.

4 Pilot Case Definitions

Family	Base case	One-at-a-time variations
American put	$T = 1, r = 0.05,$ $q = 0.02, \sigma = 0.20,$ $S_{\max} = 4,$ $M = N = 120.$	$\sigma = 0.40, r = 0.03, q = 0.00,$ plus $M = N = 180$ base check.
Dividend-paying American call	$T = 1, r = 0.05,$ $q = 0.08, \sigma = 0.20,$ $S_{\max} = 4,$ $M = N = 120.$	$\sigma = 0.40, r = 0.03, q = 0.03,$ $q = 0.10,$ plus $M = N = 180$ base check.

This is intentionally not a broad parameter sweep. It is a one-at-a-time pilot that a human reviewer can inspect.

5 Output Definitions

Output	Meaning
Value grid $U(S, \tau)$	American option value from baseline CN/PSOR.
Payoff $\Phi(S)$	Immediate exercise payoff.
Continuation premium $U - \Phi$	Diagnostic amount by which the value exceeds payoff.
Exercise indicator	1 where $U - \Phi \leq 10^{-6}$, otherwise 0.
Boundary curve	Threshold-based extraction from continuation premium.
LCP diagnostics	Obstacle, equation, and complementarity checks from Ticket 08 logic.
Delta/Gamma diagnostics	Finite-difference Greeks with boundary, kink, maturity, and strict masks.

6 Diagnostic Requirements

Every run records solver, grid, parameter, boundary, and diagnostic metadata. The acceptance rule for pilot review is conservative:

$$\text{PSOR converged, } \max(\Phi - U, 0) \leq 10^{-8}, \quad \max(b - AU, 0) \leq 10^{-6}, \quad \max |(U - \Phi)(AU - b)| \leq 10^{-6}.$$

All 11 Pilot 01 runs met this rule and were marked **ACCEPTABLE_FOR_PILOT_REVIEW**. This does not make the figures production risk surfaces. It means the small pilot is numerically coherent enough for human review.

7 Tables and Interpretation

Observed item	American put base	Dividend call base	Interpretation
PSOR convergence	True	True	All time steps converged.
Max obstacle violation	0	0	Payoff obstacle respected.
Max equation violation	$1.55664789392 \times 10^{-9}$	$2.14416621448 \times 10^{-9}$	Near diagnostic tolerance.
Max complementarity product	$6.6483476629 \times 10^{-11}$	$2.37960951462 \times 10^{-11}$	Small LCP product residual.
Boundary found rows	120	120	Boundaries found for all positive- τ rows.
Strict max $ \Gamma $	3.18001886582	2.6798873723	Greek diagnostics remain cautionary.

Higher-grid check	$M = N = 120$ boundary range	$M = N = 180$ boundary range	Reading
American put base	0.766754 to 0.934082	0.777814 to 0.955653	Boundary estimates shift with grid and remain approximate.
Dividend call base	1.066165 to 1.266630	1.044356 to 1.266615	Higher grid is a check, not a final convergence proof.

The generated tables are:

- `results/02_pilot_stress_maps/tables/pilot_01_run_summary.csv`;
- `results/02_pilot_stress_maps/tables/pilot_01_diagnostic_summary.csv`;
- `results/02_pilot_stress_maps/tables/pilot_01_boundary_summary.csv`;
- `results/02_pilot_stress_maps/tables/pilot_01_selected_slices.csv`;
- `results/02_pilot_stress_maps/tables/pilot_01_output_manifest.csv`.

8 Figures and Interpretation

Figure 1 shows option values over the main interpretation region. These are the most direct pilot stress maps, but they should be read together with diagnostics rather than as standalone production surfaces.

Figure 2 shows continuation premium $U - \Phi$. The premium is central to obstacle interpretation and to the boundary extraction, but it is not the same thing as American-minus-European value.

Figure 3 shows the exercise/continuation indicator with boundary overlay. The map is threshold-based: it visualizes the diagnostic rule $U - \Phi \leq 10^{-6}$, not an exact analytical exercise region.

Figure 4 gives selected value slices. These are easier for a greenhand reader to inspect than heatmaps because each curve can be compared with payoff.

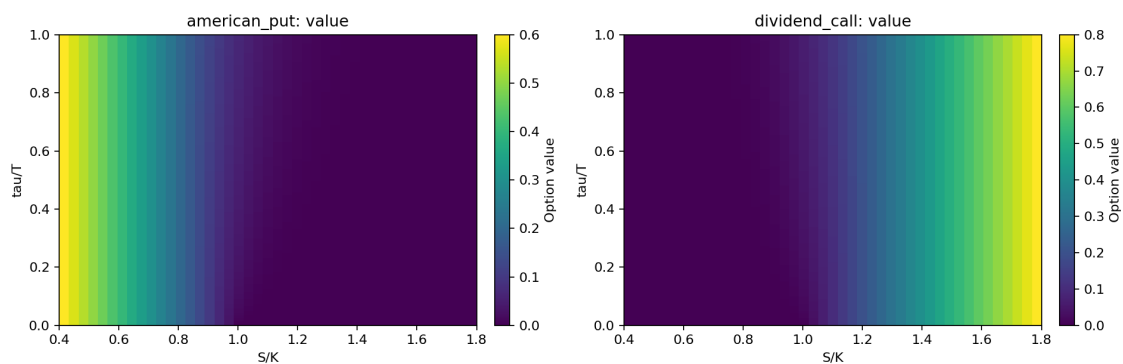


Figure 1: Base American put and dividend-paying American call value heatmaps.

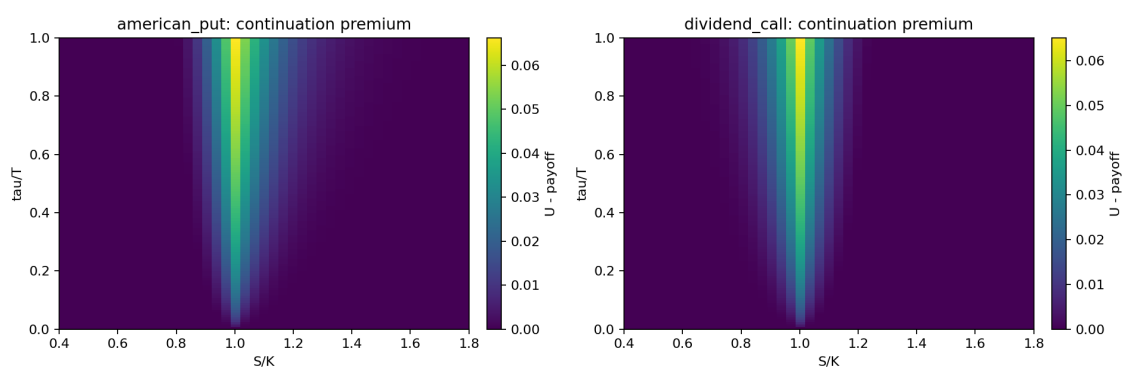


Figure 2: Base continuation-premium heatmaps.

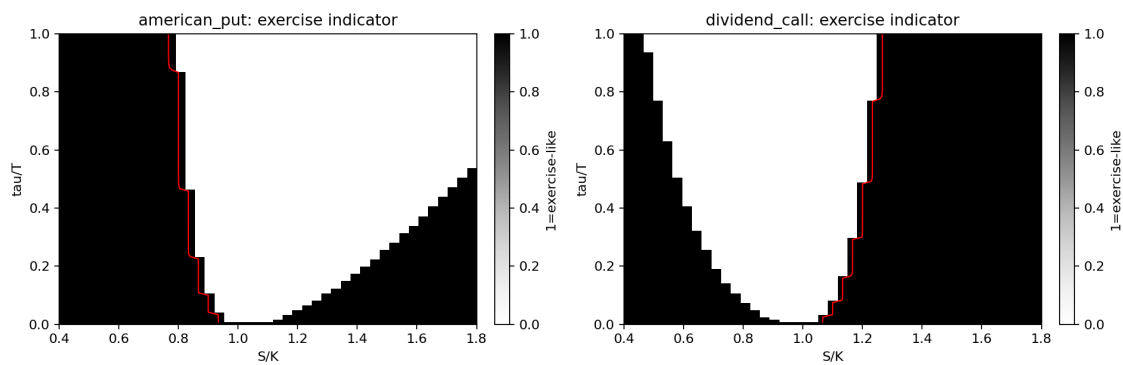


Figure 3: Exercise/continuation indicator maps with approximate boundary overlays.

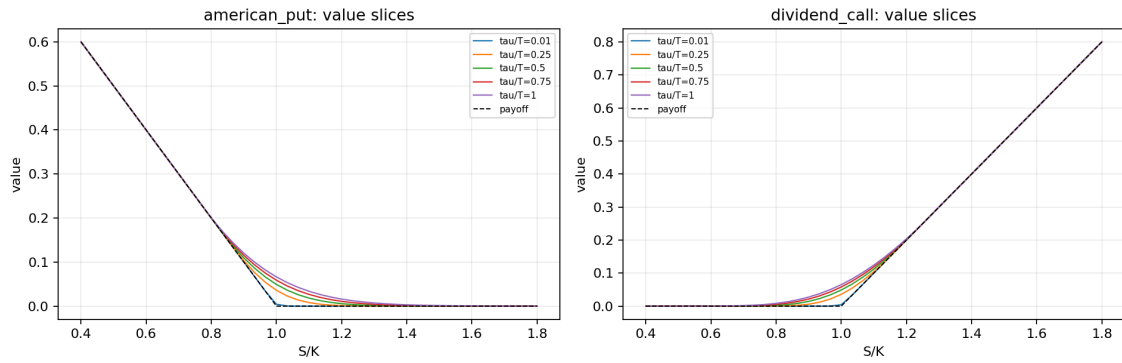


Figure 4: Selected value slices at $\tau/T = 0.01, 0.25, 0.50, 0.75, 1.00$.

Figure 5 compares boundary curves across one-at-a-time variations. This is the most useful pilot figure for seeing how dividend yield, interest rate, and volatility move the threshold boundary. It remains a diagnostic boundary curve, not an exact free-boundary claim.

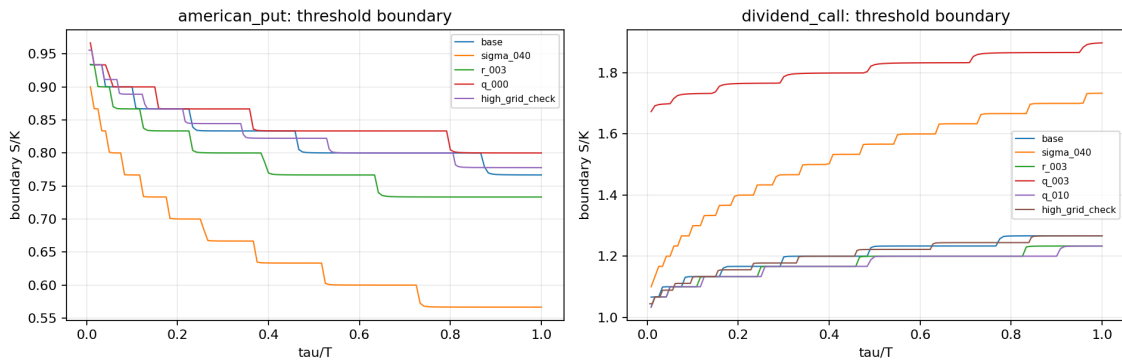


Figure 5: Approximate boundary curves across base and one-at-a-time variations.

Figure 6 shows mid-time continuation-premium slices across variations. It helps connect the boundary curves to the underlying premium quantity.

Figure 7 shows Gamma diagnostics for the two base cases. The plotted Gamma curves include caution markers for payoff kinks and extracted boundary context. They should not be used as production Greek labels.

9 Acceptance Criteria Results

Criterion	Result	Interpretation
All PSOR steps converge or report non-convergence	PASS	All 11 runs converged.
Obstacle violation near zero	PASS	All runs report maximum obstacle violation 0.
LCP diagnostics near tolerance	PASS	Largest equation violation is $1.93971868349 \times 10^{-8}$; complementarity remains small.

Criterion	Result	Interpretation
Boundary status recorded	PASS	Every run records found/not-found counts and boundary status.
Greek masks recorded	PASS	Selected slices and diagnostic summaries include masks and strict metrics.
Full parameter metadata present	PASS	Every output table records case parameters and grid metadata.
No downstream data products created	PASS	Outputs are CSV summaries and figures only.

10 Limitations

- The pilot uses a small one-at-a-time case set. It is not a broad stress-map sweep.
- Boundary extraction uses a fixed 10^{-6} continuation-premium threshold.
- The $M = N = 180$ checks are confirmation checks, not a formal convergence proof.
- Gamma remains fragile near payoff kinks, maturity rows, and boundary bands.
- Figures are pilot diagnostics. They are not production risk surfaces.
- No output should be treated as a training target for a neural surrogate.

11 What This Pilot Supports

The pilot supports a narrow claim: the validated baseline solver can generate small, metadata-rich diagnostic stress-map figures for representative American put and dividend-paying American call cases. It also supports using the same table/manifest structure for a future human-reviewed pilot expansion.

12 What This Pilot Does Not Support

The pilot does not support large-scale data generation, production Greek reporting, neural surrogate training, exact analytical free-boundary claims, or final paper-style claims about broad parameter regimes. Those steps require a separate approved plan and further validation.

13 Recommended Next Step

The recommended next step is human review of the Pilot 01 report, figures, and manifest. If the review accepts the scope and caveats, the next implementation should still be modest: a slightly expanded pilot with the same metadata contract, not a broad dataset or neural surrogate stage.

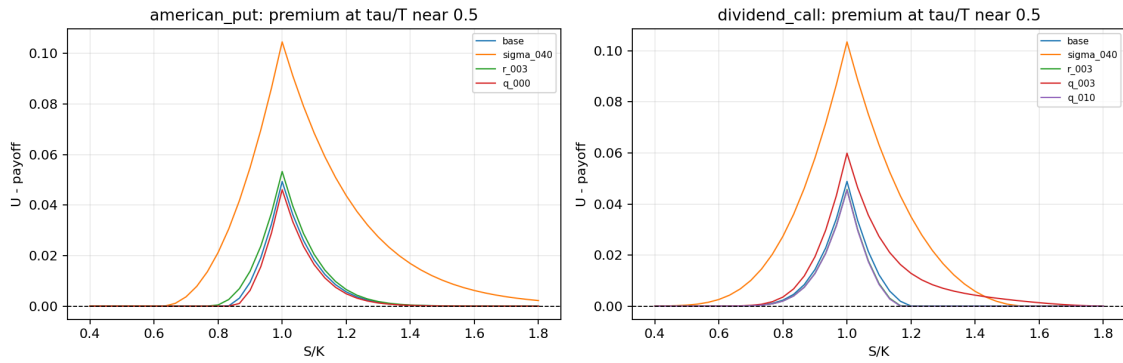
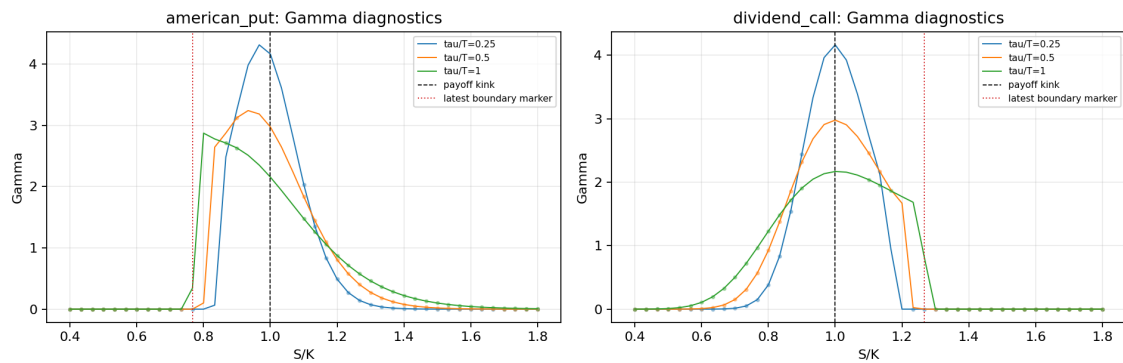
Figure 6: Continuation-premium slices across variations near $\tau/T = 0.5$.

Figure 7: Base Gamma diagnostic slices with kink and boundary cautions.

14 Reproducibility Record

Item	Record
Source module	src/american_risk_surfaces/downstream/pilot_stress_maps.py.
Experiment script	experiments/11_pilot_stress_maps.py.
Test file	tests/test_pilot_stress_maps.py.
Report source	reports/04_downstream/pilot_stress_maps_report.tex.
Report PDF	reports/04_downstream/pilot_stress_maps_report.pdf.
Output folder	results/02_pilot_stress_maps/.
Bibliography	reports/03_solver/references.bib.

References

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